

DFC REQUIREMENTS

Principle	Description
Simplicity and standardisation	Minimise complexity, use standard components, and avoid proprietary or hard-to-source parts
Manufacturing process alignment	Design PCBs and assemblies compatible with local manufacturing capabilities
Testability	Integrate test points, diagnostic interfaces, and validation hooks from the outset
Thermal and EMI/EMC design	Optimise layouts for heat dissipation and electromagnetic compliance
Documentation and traceability	Maintain detailed schematics, BOMs, and revision histories
Compliance planning	Embed regulatory and certification requirements into the design phase
Lifecycle considerations	Plan for upgrades, maintenance, and end-of-life management

and validation is still maturing, with long lead times and regulatory hurdles.

India has proven itself a global software powerhouse. However, many software engineers entering embedded domains are not trained in systematic hardware–software co-design. Startups and SMEs, while nimble, often lack the luxury of time or structured mentorship.

As a result, firmware engineers are often pushed into projects with a “code-first” mindset—writing code without proper architecture, documentation, or planning. This approach can quickly lead to unscalable, untestable, and difficult-to-debug codebases.

Go-to-market pressures: Hidden risks for startups and SMEs

Startups and SMEs racing to launch products often take shortcuts—replicating reference designs without adaptation, skipping critical reliability tests, or using educational platforms like Arduino or Raspberry Pi in production. While fast, these choices lead to immature, failure-prone products, damaging brand credibility and eroding market trust.

A common and costly mistake is selecting the wrong development toolchain early on. Developers may later realise their tools carry restrictive licences, or lack support for production or certification—forcing last-minute rewrites that delay launches and inflate costs. In high-stakes domains like automotive, this can mean redoing entire codebases.

Choosing a toolchain should go beyond features—it must account for licensing, update cycles, and long-term sustainability.

DFM best practices: Technical methodology

DFM requires a structured approach, starting with thorough requirement capture and extending through to compliance and certification. The first step involves requirements engineering, where teams define the product’s functional features, including the required interfaces and communication protocols. Environmental constraints such as temperature, humidity, vibration, altitude, and ingress protection (IP ratings) must be considered. Safety and regulatory compliance (for example, IEC, UL, EMI/EMC, RoHS, and ATEX/IECEx for hazardous environments) also need to be clearly understood. Lifecycle planning is critical to account for component longevity, upgrade paths, and overall maintainability.

The system architecture phase translates requirements into a functional design, using block diagrams to break down subsystems such as compute, power, communications, sensors, and interfaces. Flowcharts and state machines help define software behaviour, including event handling and fault management. Security must be integral from the start, incorporating secure boot, key storage, encrypted communications, and robust authentication protocols.

In India’s electronics design landscape, two priorities stand

out—delivering real customer value and ensuring design sustainability. Many teams lose time in avoidable rework due to poor upfront planning. Thoughtful design extends beyond function; it lays the groundwork for scalable and efficient product development. The key is to design with mass manufacturing in mind from the start—incorporating test hooks, debugging interfaces, and field diagnostics early on to avoid delays and cost overruns later.

Prototyping and validation are essential to reduce risks early in the development cycle. Prototypes using evaluation boards or development kits help validate core concepts before investing in custom hardware. Peer reviews of schematics and PCB layouts help identify design flaws and improve quality. Mechanical fit and clearance issues are addressed through 3D modelling using electronic design automation (EDA) tools, which also aid in thermal path analysis.

During the PCB and mechanical design stage, layouts must support DFM principles by optimising trace routing, thermal management, and minimising EMI/EMC issues. Component placement should facilitate automated soldering, inspection, and possible rework. Mechanical design must align with enclosure constraints, mounting provisions, and withstand environmental stressors.

Testability and production readiness are enhanced by embedding test points and interfaces for in-circuit testing, firmware flashing, and diagnostic access. It is important to design for remote debugging, crash logging, and field upgrades to simplify post-deployment support. Meticulous documentation—including design files, test procedures, and traceability records—is crucial for efficient manufacturing and long-term support.

Finally, compliance and certification processes should begin early, with pre-compliance testing